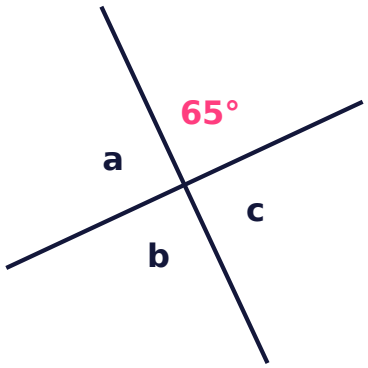
GEOMETRY QUEST · WORLD 4

Angle Rules

ANSWER SHEET · for the grown-up

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

1 Two straight lines cross. Find a, b and c. Give a reason for each. (Not drawn to scale.)

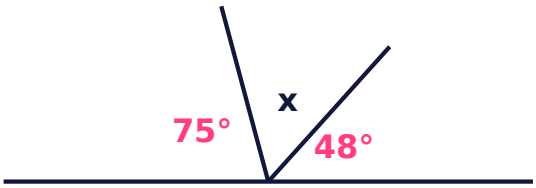


a = 115° (angles on a straight line)

b = 65° (vertically opposite)

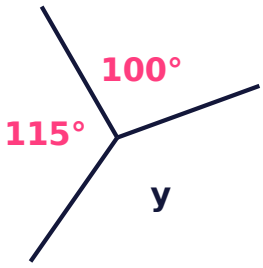
c = 115° (vertically opposite a)

2 These angles sit on a straight line. Find x.



x = 57 °

3 These angles meet at a point. Find y.



y = 145 °

4 Four equal angles meet at a point. How big is each one?

90 °

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5 Open a pair of scissors. Explain why the angle between the blades always equals the angle between the handles.

Two straight lines cross, so vertically opposite angles are equal.

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